

LabProgram2 – Conditional Statements Report

1. Aim

To implement basic conditional statements in Java through four problems: checking whether a number is even or odd, finding the largest of three numbers, assigning a grade from a percentage, and displaying a day of the week using a switch statement.

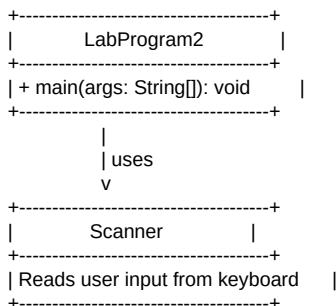
2. Description

The program is written in Java under the package `observationtasks` and uses the `Scanner` class to receive keyboard input. The source contains four separate practice problems, with the first three placed inside comments and the fourth problem active. The first problem uses if-else, the second uses if-else-if conditions with logical AND, the third assigns grades using percentage ranges, and the fourth uses switch-case to display the day corresponding to a number.

3. Tools Used

- Programming Language: Java
- Development Environment: Any Java IDE or text editor
- Execution Tool: Java Development Kit (JDK)
- Input Class: `java.util.Scanner`
- Concepts: if-else, else-if ladder, logical operators, switch-case, break, and user input

4. UML Diagram



Control structures used:

- if / else-if / else
- switch-case
- break
- relational and logical operators

5. Algorithm

1. Start the program.
2. Import the `Scanner` class.
3. Create a `Scanner` object to read input.
4. For Problem 1, read a number and check whether `number % 2` equals zero. Display even or odd.
5. For Problem 2, read three numbers and compare them using if-else-if conditions. Display the largest number.
6. For Problem 3, read the percentage and check the ranges from 90 down to 60. Display the corresponding grade or FAIL.
7. For Problem 4, read a day number and use switch-case to display the day name.

8. If the day number does not match a case, display invalid day.
9. Stop the program.

6. Complete Java Source Code

```
package observationtasks;
import java.util.Scanner;

/* Problem 1: Check whether a number is even or odd
public class LabProgram2 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n;
        System.out.println("enter the number");
        n = sc.nextInt();

        if (n % 2 == 0) {
            System.out.println("the number is even");
        } else {
            System.out.println("the number is odd");
        }
    }
}
*/

/* Problem 2: Find the largest of three numbers
public class LabProgram2 {
    public static void main(String[] args) {
        Scanner max = new Scanner(System.in);
        int a, b, c;

        System.out.println("enter the values of a,b,c");
        a = max.nextInt();
        b = max.nextInt();
        c = max.nextInt();

        if (a >= b && a >= c) {
            System.out.println("the largest number is " + a);
        } else if (b >= a && b >= c) {
            System.out.println("the largest number is " + b);
        } else {
            System.out.println("the largest number is " + c);
        }
    }
}
*/

/* Problem 3: Display grade based on percentage
public class LabProgram2 {
    public static void main(String[] args) {
        Scanner p = new Scanner(System.in);
        double percentage;

        System.out.println("enter the percentage");
        percentage = p.nextDouble();

        if (percentage >= 90)
            System.out.println("GRADE - A");
        else if (percentage >= 80)
            System.out.println("GRADE - B");
        else if (percentage >= 70)
            System.out.println("GRADE - C");
        else if (percentage >= 60)
            System.out.println("GRADE - D");
        else
            System.out.println("FAIL");
    }
}
*/

/* Problem 4: Display day of the week
Active program
*/
public class LabProgram2 {
    public static void main(String[] args) {
        Scanner d = new Scanner(System.in);
        int day;

        System.out.println("enter the day");
        day = d.nextInt();

        switch (day) {
            case 1:
                System.out.println("sunday");
                break;
            case 2:
```

```
        System.out.println("monday");
        break;
    case 3:
        System.out.println("tuesday");
        break;
    case 4:
        System.out.println("wednesday");
        break;
    case 5:
        System.out.println("thrusday");
        break;
    case 6:
        System.out.println("friday");
        break;
    default:
        System.out.println("invalid day");
        break;
    }
}
```

7. Sample Output

Problem 1: Even or Odd

Input:

enter the number

8

Output:

the number is even

Problem 2: Largest of Three Numbers

Input:

enter the values of a,b,c

12 25 18

Output:

the largest number is 25

Problem 3: Grade Percentage

Input:

enter the percentage

86

Output:

GRADE - B

Problem 4: Day of the Week (Active Program)

Input:

enter the day

3

Output:

tuesday

For an invalid day:

Input:

enter the day

7

Output:

invalid day

8. Result

The four conditional-statement problems were implemented in Java. The programs demonstrate decision-making using if-else, else-if conditions, logical operators, and switch-case. The active fourth program displays the day of the week for valid inputs from 1 to 6 and displays an invalid-day message for other inputs.